

FIN 4000 Midterm Study Guide

Coverage: BKM Chapters 1-5 and 14-16 · **Updated:** October 6, 2026 · **Instructor:** Jim Northey

Overview

The midterm will have a set of multiple-choice questions in CFA format and a set of calculations, with at least one calculation from each chapter. The exam is timed so that students who have studied should finish in less than an hour.

- The exam is on paper and taken in class.
- You may use a TI BA II Plus or TI BA II Plus Professional calculator. Smartphone calculator apps are not permitted.
- A practice exam will be available by Monday, October 12.

Each chapter below lists the topics to know, the study guide that covers each one, and the BKM reading, followed by what you should be able to do.

Chapter 1 — The Investment Environment

Topic	Study guide	BKM reading
Real vs. financial assets	mini-01-real-vs-financial-assets	Ch. 1.1
Intangible assets	mini-17-intangible-assets	Ch. 1.1

You should be able to:

- Determine whether an asset is an intangible asset.
- Explain the difference between real and financial assets.

Chapter 2 — Asset Classes and Financial Instruments

Topic	Study guide	BKM reading
Municipal bonds	—	Ch. 2.2, Equations 2.1 and 2.2
Day count conventions	—	Ch. 14.2

You should be able to:

- Calculate the tax-equivalent yield to compare a taxable bond with a municipal bond.

Chapter 3 — How Securities Are Traded

Topic	Study guide	BKM reading
Purchasing stock on margin	mini-02-short-sales-and-margin-calls	Ch. 3.8
Short selling stock	mini-02-short-sales-and-margin-calls	Ch. 3.9
How Treasuries are issued	mini-11-bond-issuance-underwriting-and-oid	—
How initial public offerings work	gen-09-ipo-process-and-underwriters-role	—

You should be able to:

- Calculate the rate of return from buying stock on margin.
- Determine the price at which you will receive a margin call.
- Calculate the return from selling stock short.

Chapter 4 — Mutual Funds and Other Investment Companies

Topic	Study guide	BKM reading
Net asset value	supp-01-nav-and-etf-arbitrage	Ch. 4.1
Fund costs: front-end and back-end loads, no-load funds, expense ratio	mini-03-mutual-fund-costs	Ch. 4.4
Closed-end and open-end funds	—	—
Tax differences between ETFs and mutual funds	mini-14-etf-in-kind-creation-redemption	—
How Approved	supp-01-nav-and-etf-arbitrage	—

Chapter 5 — Risk, Return, and the Historical Record

Topic	Study guide	BKM reading
Geometric vs. arithmetic mean	mini-04-arithmetic-vs-geometric-returns	Ch. 5.6
Holding-period returns	—	Ch. 5.3
Risk-free rate, risk premiums, and the Sharpe ratio	gen-10-sharpe-ratio-and-risk-premiums	Ch. 5.3

You should be able to:

- Describe the historical record by asset class.
- Calculate a holding-period return.
- Calculate a risk premium from historical returns, and judge investments by comparing Sharpe ratios.

Chapter 14 — Bond Prices and Yields

Topic	Study guide	BKM reading
U.S. Treasuries	gen-06-us-treasuries	Ch. 14.1
Corporate bonds	—	Ch. 14.1
Callable and putable bonds	gen-07-callable-putable-convertible-bonds	Ch. 14.1
Preferred stock	—	Ch. 14.11
Accrued interest: flat (clean) price and dirty (invoice) price	—	Ch. 14.1
Treasury STRIPS	gen-01-treasury-strips	Ch. 14.4

You should be able to:

- Determine a bond's dirty price from its quoted price.
- Match liabilities using STRIPS.

Chapter 15 — The Term Structure of Interest Rates

Topic	Study guide	BKM reading
Using forward rates to price a bond and determine its YTM	mini-16-spot-vs-forward-rates	Ch. 15.2
Spot rates vs. forward rates	mini-16-spot-vs-forward-rates	Ch. 15.2

You should be able to:

- Calculate a bond's price from forward rates, then use that price to determine its YTM.
- Calculate a bond's price from discount rates, and use it to determine whether an arbitrage opportunity exists.

Chapter 16 — Managing Bond Portfolios

Topic	Study guide	BKM reading
Convexity and duration	mini-07-convexity-and-duration-convexity	Ch. 16.1-16.2
Duration, average life, and convexity	gen-05-duration-average-life-convexity	Ch. 16.1-16.2

You should be able to:

- Compare bonds by their duration and convexity.
- Explain the risks involved in liability matching.

Current Events

Topic	Study guide	BKM reading
U.S. Treasury market intervention and its impact	ce-05-treasury-buybacks-and-bond-yields	Ch. 16.1
Why bond yields are rising	ce-03-why-bond-yields-are-rising	—
Yen carry trade	ce-04-yen-carry-trade-and-treasury-roles	—

You should be able to:

- Explain the impact of Treasury buybacks.
- Explain how rising bond yields have affected the yen carry trade.